# Alzheimer progression prediction: project roadmap

ADNI + OASIS-2, multimodal and longitudinal. Status on 2 November 2026

**2 done** | **3 ready to run** | **3 to do**

# Eight phases, three states

1 DONE

## Scope and data inventory

Folders, file types, subjects

2 DONE

## Data reduction and checks

27 GB to 6 GB, all checks pass

3 READY TO RUN

## Tables and labels

Features by modality, 36-month label

4 READY TO RUN

## Image preprocessing

Every scan to a $96^3$ array

5 READY TO RUN

## Baseline experiments

RQ1 to RQ3 on tabular data

6 TO DO

## 3D-CNN + BRNN model

Images and visit sequences

7 TO DO

## Spark Big Data pipeline

Required by the project brief

8 TO DO

## Report and defense

Methods, results, limits

Done

Script written, run on your data

Still to build

# Done: data understood and reduced

**27 GB to 6 GB**
folder size after reduction

**ADNI: 62 subjects**
124 MRI visits, 21 with one visit

**OASIS-2: 82 subjects**
209 MRI visits, 2 to 5 each

## What we found

- No OASIS-3 and no PET images on disk
- ADNI tables cover clinical scores, CSF, plasma, PET summaries
- One scan per ADNI visit, 3 to 4 repeats per OASIS-2 session

## Checks passed

- Files, IDs, class balance, table coverage
- Images load, one shape per cohort
- Empty 2019 ADNI scan skipped automatically

# Ready to run: three scripts

### build_tables.py

One row per MRI visit, features by modality, 36-month progression label.

```
python build_tables.py
```

**Send: tables_report.txt**

### preprocess_images.py

Every scan to a $96^3$ array, same orientation in both cohorts.

```
python preprocess_images.py
--limit 6
```

**Check: qc_preprocessed.png**

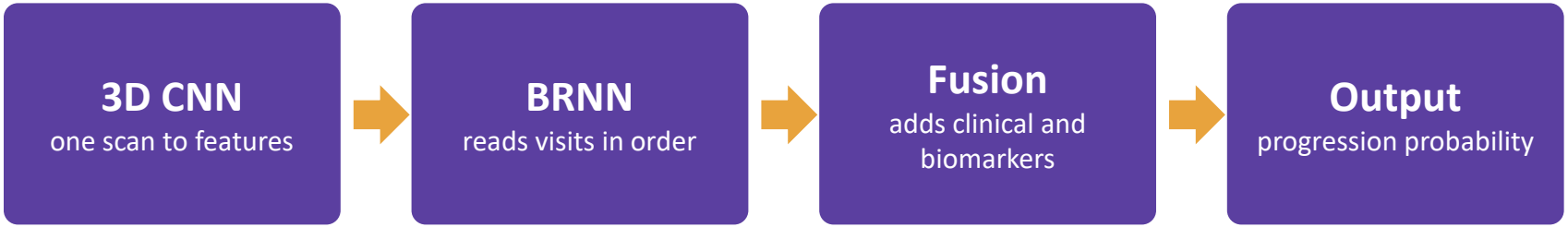
### run_experiments.py

RQ1 to RQ3 baselines with subject-level cross-validation.

```
python run_experiments.py
```

**Send: summary.txt**

# To do: the deep model




## Experiments it must answer

**RQ1** Images only vs tables only vs both

**RQ2** One visit vs the full visit sequence

**RQ3** Drop a modality or a visit at test time

# To do: Big Data pipeline and report

## Spark pipeline

- Parallel feature extraction over scans and tables
- Distributed training, timing vs workers and data size
- Start from oasis2_pipeline.zip

## Report and defense

- Methods: cohorts, labels, leakage controls
- Results: RQ1 to RQ3 tables and plots
- Limits: small sample, no PET images, CDR-based OASIS-2 labels
- Cite the five references from the brief

# Questions, answers and risks

| Research question | Baseline (script ready) | Deep model (to do) |
|---|---|---|
| RQ1 multimodal vs unimodal | Feature groups alone vs combined | Images only vs images + tables |
| RQ2 longitudinal vs static | Static features vs added history | One visit vs visit sequence |
| RQ3 robustness | Mask modalities and values at test time | Drop scans, visits, modalities |

**Risks** About 100 labeled visits, so compare ± ranges, not rankings. OASIS-2 labels come from CDR and differ from ADNI diagnoses. No PET images or OASIS-3 locally.